

RECEP TALHA YALÇINKAYA

COMPUTER ENGINEER

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EDUCATION

2022 - 2026

PAMUKKALE UNIVERSITY, DENIZLI

- B.S. in Computer Engineering
- GPA: 3.08 / 4.00

2018 - 2022

60. YIL ANATOLIAN HIGH
SCHOOL, IZMIR

TECHNICAL SKILLS

- Languages: Python, C++, C#, MATLAB
- AI & Vision: PyTorch, OpenCV, YOLO, Deep Learning, Machine Learning
- Automotive AI: ADAS, Sensor Fusion
- Edge & Embedded: NVIDIA Jetson, Raspberry Pi, Linux, CUDA, CAN BUS

SOFT SKILLS

- "PAU ATAY" Team ADAS & Autonomous & Software Leader
- Project & Task Management
- Analytical Problem Solving
- Disciplined & Result-Oriented

LANGUAGES

- English (Intermediate)

PROFILE

Results-driven Autonomous Systems & Edge AI Engineer with a strong foundation in developing real-time ADAS solutions and sensor fusion algorithms. Adept at bridging the gap between high-performance software and embedded hardware, deploying complex computer vision and deep learning models onto edge devices like NVIDIA Jetson and Raspberry Pi. Recognized as a TEKNOFEST Finalist and certified in Autonomous Driving Technologies by the Ministry of Industry and Technology, passionate about building scalable, intelligent vehicle architectures.

EXPERIENCE & PROJECTS

Robust Traffic Sign Recognition (B.S. Thesis) SEP 2025 - JUN 2026

Computer Vision Developer

- ACHIEVEMENT:** My thesis project, running on Raspberry Pi 5 and Hailo AI hardware, won **1st Place** in the Artificial Intelligence Category among Pamukkale University Computer Engineering thesis projects.
- Developed a computer vision (CV) processing pipeline to classify "morphological siblings" on speed limit signs, reliably distinguishing between 80 km/h and 30/60/90 km/h variants.
- Resolved critical edge cases while maintaining high detection accuracy even when structurally critical nodes were damaged, obstructed, or contaminated.

Senkron R&D (PAU Technopark)

JUN 2025 - JUL 2025

ADAS Software Engineering Intern

- Assisted in the R&D of Advanced Driver Assistance Systems (ADAS), focusing on algorithm testing and optimization. Gained practical experience in vehicle software architecture, real-time data processing, and sensor integration.

ADAS Track Navigation System (TEKNOFEST)

2024 - 2025

ADAS & Embedded Systems & Telemetry Developer

- Developed a real-time track navigation dashboard for the Efficiency Challenge Electric Vehicle using Python, deployed on a Linux Ubuntu-based Raspberry Pi 5.
- Integrated a UBLOX NEO-M8N GPS module to extract live satellite coordinates, dynamically plotting the vehicle's position on a custom parsed KML track map (TÜBİTAK campus).
- Engineered a Euclidean distance-based map-matching algorithm to snap the live vehicle symbol accurately to the track, alongside automated lap counting and lap time calculation logic.

CERTIFICATES

Autonomous Driving Technologies

National Technology Academy

Date: February 2026

ID: 70964848966370

TEKNOFEST Finalist

TÜBİTAK - Efficiency Challenge

Date: September 2025

ID: 78787083686857